

ACCESS Application Testing & System Specification Report

Final Year Project (PSM) Development, Multi-User Feature Matrix, WCAG 2.2 Accessibility Engine, Quality Assurance & Verification Benchmark

EXECUTIVE SUMMARY & PURPOSE: This comprehensive document serves as an evaluation, self-testing, and quality-assurance benchmark for the **ACCESS (Accessible Cognitive Education & Support System)** platform. It outlines the three distinct user portals (Administrator, Educator, Learner), details the WCAG 2.1/2.2-compliant accessibility engine with neurodivergent presets, details recent architectural updates (rich learning progress telemetry, sensory calm mode, Esc key shortcut, preset info tooltips, side-by-side certificate badges), details educator compliance audit rules, defines automated testing with Playwright and k6, and summarizes the complete seeded database architecture.

1. User Role Capabilities & Functional Architecture

The ACCESS platform implements strict multi-tiered Role-Based Access Control (RBAC) across three distinct user roles. Each role is equipped with dedicated interfaces, telemetry pipelines, and accessibility safeguards:

1.1 Administrator Portal (/admin)

- **Executive Command Center (/admin):** Real-time visual telemetry tracking total platform engagement, student enrollments, course publications, verified certificates, and active user health.
- **User Directory & Account Governance (/admin/users):** Complete oversight of all registered learner and educator accounts, role promotions, status toggling, and profile metadata management.
- **Deep Learner Telemetry Inspection (/admin/users/[id]):** Granular inspection of individual student progress, lesson completion logs, quiz attempts, checkpoint completions, and issued credentials.
- **Course Moderation & Quality Control (/admin/courses):** Review, inspect, publish, archive, or approve educator-authored courses and manage system-wide foundational curriculum.
- **Instructor Vetting & Approvals (/admin/instructor-applications):** Evaluation pipeline for educator applications with qualifications, verification documents, and CV portfolios.
- **Cryptographic Certificate Oversight (/admin/certificates):** Registry of all digital credentials issued on the platform with QR-code cryptographic UUID verification.
- **Feedback & Inquiry Center (/admin/contact-messages):** Centralized inbox triaging inbound contact submissions, learner inquiries, and accessibility feedback.
- **Automated Report Generator (/admin/reports):** PDF export engine delivering administrative executive summaries, course analytics, and accessibility compliance data.

Self-Improvement & Development Notes: ADMINISTRATOR PORTAL

[Use this space to document your specific additions, bug fixes, UI enhancements, and feature refinements for this user tier]

- **Improvement 1:** _____
- **Improvement 2:** _____
- **Improvement 3:** _____
- **Tested Status:** [☐] Verified on Localhost [☐] Verified on Vercel Production [☐] Database Synced

1.2 Educator Portal (/educator)

- **Course Studio & Builder (/educator/courses):** Modular course authoring interface supporting multi-chapter hierarchies, rich lessons, interactive activities, and video materials.
- **Rich TipTap Content Editor:** Accessible WYSIWYG editor supporting accessible typography, custom callout cards, semantic tables, image alignment, and direct media integration.

- **Automated Accessibility Compliance Engine:** Real-time deterministic evaluator scanning lesson markup, computing a 0-100 score against WCAG/COGA rules, and providing 1-click automatic remediation.
- **Image Descriptions & Alt Text Tool:** Embedded media manager extracting all lesson images, flagging missing descriptions, and enabling inline alt-text authoring with immediate markup binding.
- **Interactive Video Checkpoints:** In-video multiple-choice question authoring tool that binds questions to exact timestamps, pausing playback automatically for active recall.
- **Native Interactive Activities:** Authoring suite for 5 interactive exercise types: Flashcards, Drag & Drop, Fill in the Blanks, Memory Match Game, and Chronological Timeline.
- **Student Cohort Intelligence (/educator/students):** Progress tracking across enrolled students with completion percentages, quiz scores, checkpoint completions, and engagement metrics.
- **Course Performance Analytics (/educator/analytics):** Telemetry visualizers displaying drop-off detection, quiz performance distributions, and lesson engagement durations.
- **Custom Certificate Designer (/educator/certificates):** Course-specific credential designer allowing custom signatures, institutional titles, criteria, and design templates.

Self-Improvement & Development Notes: EDUCATOR PORTAL

[Use this space to document your specific additions, bug fixes, UI enhancements, and feature refinements for this user tier]

- **Improvement 1:** _____
- **Improvement 2:** _____
- **Improvement 3:** _____
- **Tested Status:** ☐ Verified on Localhost ☐ Verified on Vercel Production ☐ Database Synced

1.3 Learner Portal (/learner)

- **Adaptive Learning Workspace (/learner/courses/[id]):** Highly flexible lesson viewer supporting Standard View, Slideshow/Paginated View, and Micro-chunked sections.
- **One-Click Accessibility Presets:** Instant application of evidence-based profiles: Dyslexia, ADHD, Autism & Sensory, Low Vision, and Custom configurations.
- **Assistive Text-to-Speech (TTS) Engine:** Multi-speed browser speech synthesis with **Shift-Key Cursor Inspection** (reads text, image alt text, and interactive buttons) and full-lesson audio narration.
- **Executive Function Support Suite:** Visual Step-by-Step Task Checklist, Progress Timeline with status milestones, NowBar current task tracker, and Auto-Save reassurance indicators.
- **Sensory & Cognitive Load Reducers:** Reading Spotlight focus ruler (viewport-center tracking), Distraction-Free full-screen mode with **Esc Key Shortcut**, Soft Background Tints (Irlen filters), and Zero-Animation mode.
- **Detailed Learning Progress Dashboard (/learner/progress):** 5 Granular telemetry cards (Active Study Time, Total Visits, Lessons Done, Quiz Mastery, Streaks), 4 tabs (Overview, Courses, Visit History, Quiz Tries), and Sensory Calm mode.
- **Interactive Video Player:** Dedicated endpoint (/api/lessons/[id]/video-questions) with interactive timestamp checkpoint badges that seek playback and trigger question overlays.
- **Side-by-Side Dual Digital Credentials:** Course detail page displaying both Institutional Platform Certificates and Educator Custom Certificates in a responsive 2-column grid.

Self-Improvement & Development Notes: LEARNER PORTAL

[Use this space to document your specific additions, bug fixes, UI enhancements, and feature refinements for this user tier]

- **Improvement 1:** _____
- **Improvement 2:** _____
- **Improvement 3:** _____
- **Tested Status:** ☐ Verified on Localhost ☐ Verified on Vercel Production ☐ Database Synced

2. In-Depth Accessibility Architecture, Presets & WCAG 2.2 Standards

ACCESS operationalizes the **Web Content Accessibility Guidelines (WCAG 2.2 Level A/AA/AAA)** and **W3C Cognitive Accessibility (COGA)** standards. Accessibility is structured into distinct evidence-based presets designed around specific neurodivergent learning profiles:

Preset & Target Profile	Applied Technical Adaptations	WCAG 2.2 / COGA Criteria	Cognitive & Scientific Rationale (Why it helps)
Dyslexia Preset Target: Learners with phonological processing difficulties, visual crowding, or reading fatigue.	• Atkinson Hyperlegible / OpenDyslexic font • Font size 19px, line-height 1.7x • Word spacing +40% (0.16em floor) • Cream background tint (#FFFDD0) • Reading Spotlight focus ruler • Text-to-Speech with Shift image alt reading	• WCAG 1.4.12 Text Spacing (AA) • WCAG 1.4.8 Visual Presentation (AAA) • WCAG 1.1.1 Non-text Content (A) • WCAG 1.4.5 Images of Text (AA) • BDA Dyslexia Style Guide	Dyslexic readers frequently experience 'visual crowding' where adjacent letterforms blur together. Increased letter/word spacing and specialized character weights prevent optical confusion (b/d/p/q). Cream tint reduces glare-induced visual stress (Meares-Irlen syndrome).
ADHD Preset Target: Learners with executive dysfunction, attention deficit, and working memory constraints.	• Action-First Priority Banner (immediate 1-click resume) • Chunked Micro-learning (H2 sections / Tabs) • Distraction-Free mode with Esc key exit • Step-by-Step Task Checklist • Visual Progress Timeline • In-video question checkpoints (≤360s) • Auto-Save indicator & Soft Grey tint	• W3C COGA 4.2 Chunk Information • W3C COGA 5.1 Short Sections • W3C COGA Objective 6 (No memory rely) • WCAG 2.2.4 Interruptions (AAA) • WCAG 2.2.1 Timing Adjustable (A)	Working memory buffers in ADHD learners overflow quickly when processing unbroken blocks of text. Breaking lessons into bite-sized chunks prevents cognitive overwhelm. Active in-video checkpoints maintain dopamine-driven focus, and auto-save eliminates lost-work anxiety.
Autism & Sensory Preset Target: Learners with sensory hypersensitivity, cognitive rigidity, or anxiety regarding unpredictability.	• Sensory Calm Progress Mode (collapses dense numeric stats) • Continuous predictable scroll layout • Zero animation (animation_level='none') • Muted pastel color palette (#E0F2FE) • Clear Learning Objectives upfront (≥2) • Plain Language Summary Card	• WCAG 2.3.3 Animation from Interaction (AAA) • WCAG 3.2.3 Consistent Navigation (AA) • W3C COGA 3.1 Content Summary • W3C COGA 6.1 Clear Expectations • W3C COGA 2.3 Avoid Figurative Speech	Unexpected motion, dense numeric charts, or unpredictable layouts trigger sensory overload and performance anxiety. Stating plain-language narrative summaries upfront anchors comprehension while hiding dense metrics prevents numerical anxiety.
Vision & General Preset Target: Low vision, age-related vision loss, situational light constraints, or ESL learners.	• High-contrast typography (>7:1 contrast) • Resizable text scaling up to 24px • Multi-speed TTS voice narration • Shift-hover image alt description reading • Keyboard navigation with visible focus rings	• WCAG 1.4.3 Contrast (Minimum) (AA) • WCAG 1.4.6 Contrast (Enhanced) (AAA) • WCAG 2.4.7 Focus Visible (AA) • WCAG 2.1.1 Keyboard Navigation (A)	Guarantees that all content and interface controls remain fully legible and navigable without a pointing device. Screen reader alt text ensures non-sighted learners receive full parity of visual information.

3. Recent Architectural & Quality-of-Life Enhancements

The latest development iteration introduced six major usability, telemetry, and accessibility refinements:

- **Detailed Learning Progress Dashboard (/learner/progress):** Expanded telemetry incorporating Active Study Time (hours/mins), Lessons Completed vs. Total Visits (repetition log), Quiz Mastery %, and Daily Streak counters.
- **Multi-Tabbed Session & Assessment Telemetry:** Four cognitive tabs: (1) *Overview* with quick resume, (2) *Courses* with mastery gauges, (3) *Visit History* with relative timestamps ("15m ago", "Yesterday"), session durations, and visit counts, and (4) *Quiz Tries* with attempt numbers, scores, and pass/review tags.
- **Sensory Calm Mode & ADHD Priority Layout:** In Autism / Pale Blue mode, dense numeric stat boxes are collapsed into a clean toggle ([Show detailed numeric cards]) to eliminate visual crowding. In ADHD mode, the *'Resume Where You Left Off'*

banner is placed at the top for zero decision friction.

- **Preset Details Modal Tooltip Refactoring (PresetDetailsDialog.tsx):** Compacted 9-item setting diff list by replacing long inline text blocks with neat circular (i) info buttons with hoverable tooltips explaining the WCAG/COGA rationale.
- **Global Keyboard Shortcut for Distraction-Free Mode:** Added global Escape (Esc) key listener in LearnerShell.tsx allowing instantaneous exit from Distraction-Free mode, accompanied by a styled `<kbd>Esc</kbd>` badge.
- **Side-by-Side Dual Certificate Display (CourseDetailPage.tsx):** Restructured the certificate display into a responsive 2-column grid showing both the Institutional Platform Certificate (Green) and Educator Custom Certificate (Purple) side-by-side with equal-height alignment.

4. Educator Accessibility Audit Engine & Quality Checker

To ensure that accessibility is embedded at the authoring stage rather than treated as an afterthought, ACCESS includes an **automated, real-time accessibility compliance auditor** embedded directly in the Course Studio. The auditor evaluates lesson draft markup against 14 deterministic rules with zero hallucinations, no external API latency, and complete mathematical reproducibility.

4.1 The 14 Deterministic Audit Standards Across Focus Profiles

Rule ID & Standard Title	Profile / Scope	Strict Threshold / Requirement	Source Standard & Citation
base_content Teachable Content	Universal (All)	At least 50 words of body text or an embedded video.	WCAG 2.2 -- 1.3.1 Info & Relationships
base_alt_text Image Alternative Text	Universal (All)	Every <code></code> tag must carry non-empty, descriptive alt text.	WCAG 2.2 -- 1.1.1 Non-text Content (Level A)
base_headings Heading Structure	Universal (All)	Strict hierarchical sequence (H1 → H2 → H3). No skipped levels.	WCAG 2.2 -- 2.4.6 Headings & Labels (Level AA)
base_transcript Video Transcript	Universal (All)	Any lesson containing a video must provide a text transcript.	WCAG 2.2 -- 1.2.1 Audio/Video-only (Level A)
base_link_text Descriptive Link Text	Universal (All)	Flags vague anchor text ('click here', 'more', 'link').	WCAG 2.2 -- 2.4.4 Link Purpose in Context (A)
base_duration Estimated Duration	Universal (All)	Lesson reading duration must be explicitly estimated (≤45 min).	W3C COGA -- Objective 5 (Pacing)
base_spacer_paragraphs No Empty Spacers	Universal (All)	Max ≤3 empty <code><p></p></code> tags used for manual visual spacing.	WCAG 2.2 -- 1.3.1 Info & Relationships
adhd_chunking Cognitive Chunking	ADHD Profile	Section breaks (H2/HR) required every ≤400 words.	W3C COGA -- Pattern 4.2: Chunk Information
adhd_video_len Active Video Recall	ADHD Profile	Videos >6 minutes (>360s) require timestamped in-video questions.	W3C COGA -- Objective 5: Help Users Focus
asd_objectives Clear Objectives	Autism Profile	At least ≥2 explicit learning goals stated up front.	W3C COGA -- Pattern 6.1: Clear Expectations
asd_summary Simplified Summary	Autism Profile	Plain-language summary of ≥15 words provided.	W3C COGA -- Pattern 3.1: Content Summary
dys_paragraphs Paragraph Word Limit	Dyslexia Profile	Maximum 80 words per paragraph to prevent visual crowding.	British Dyslexia Association Style Guide
dys_sentences Sentence Length	Dyslexia Profile	Maximum 30 words per sentence; mean sentence length ≤20 words.	British Dyslexia Association Style Guide
dys_readability Flesch-Kincaid Score	Dyslexia Profile	Flesch-Kincaid Grade Level ≤Grade 8 for general audiences.	Kincaid et al. (1975) Readability Formula

4.2 Scoring Algorithm & Mutual Benefits for Educators and Learners

Scoring & Compliance Bands: The audit engine computes a weighted score (0-100%). Scores $\geq 80\%$ earn a **Pass / Good (Green)** status, 50-79% triggers a **Needs Attention (Amber)** warning with actionable inline remediation buttons, and $< 50\%$ indicates **Critical Issues**.

- **How it helps the Educator:** Eliminates guesswork. Instead of memorizing dense WCAG documentation, educators receive immediate visual alerts, exact line-level highlights, and 1-click automatic fixes (e.g., auto-chunking text, formatting objectives, embedding alt-text fields).
- **How it helps the Student:** Ensures that every published lesson is guaranteed to work with screen readers, assistive TTS, keyboard shortcuts, and cognitive pacing aids. No learner encounters broken markup or inaccessible media.

5. Quality Assurance & Automated Testing Plan

To validate platform reliability, role-based security guards, and high-concurrency throughput, the quality assurance plan uses two industry-standard testing tools: **Playwright** for End-to-End Functional Verification and **k6 by Grafana** for Concurrency & Load Stress Testing.

5.1 Tool 1: Playwright (End-to-End Functional & Integration Testing)

- **What it is:** A modern browser automation testing framework developed by Microsoft that automates Chromium, Firefox, and WebKit under real browser conditions.
- **What it validates:**
 1. Multi-role authentication (Admin, Educator, Learner) and RBAC route protection.
 2. Accessibility preset switches (verifies DOM styling, font changes, line spacing, and dark/soft/tint modes).
 3. Course enrollment flows, lesson slide progression, and interactive quiz submission.
 4. TTS Shift-key inspection, Esc key shortcut, and video checkpoint question triggers.
- **What to expect / Result:** Visual screenshots of test states, network trace logs, test execution time per scenario, and an interactive HTML report confirming 100% pass rate on all critical user journeys.
- **How to execute Playwright:**

```
# 1. Install Playwright and browser dependencies
npm install -D @playwright/test
npx playwright install --with-deps chromium

# 2. Run all end-to-end test scenarios against Localhost
npx playwright test

# 3. Run against Production URL with interactive UI mode
npx playwright test --ui --env BASE_URL=https://access.vercel.app

# 4. View interactive HTML test report
npx playwright show-report
```

5.2 Tool 2: k6 by Grafana (Load, Stress & High-Concurrency Testing)

- **What it is:** A high-performance load testing engine written in Go, scripted in JavaScript, designed for simulating heavy concurrent traffic.
- **What it validates:**
 1. API response latencies under sustained concurrent request loads.
 2. Supabase serverless database connection pooling limits and caching efficiency.
 3. System SLA thresholds: p95 latency $< 500\text{ms}$, error rate $< 1\%$ under peak stress.
- **What to expect / Result:** Real-time CLI output displaying total requests completed, Requests Per Second (RPS), p90/p95/p99 latency percentiles, and check pass/fail ratios.
- **How to execute k6:**

```
# 1. Install k6 on Windows
winget install k6 --source winget
```

```
# 2. Run standard load test (50 concurrent VUs over 2 minutes)
k6 run -e BASE_URL=http://localhost:3000 scripts/k6-load-test.js

# 3. Run high-concurrency stress test (500 VUs ramp-up against Production)
k6 run --vus 500 --duration 3m -e BASE_URL=https://access.vercel.app scripts/k6-load-test.js

# 4. Export test metrics to JSON summary
k6 run --summary-export=k6-summary.json scripts/k6-load-test.js
```

5.3 Concurrency Architecture: 50 Database Users vs 500 Concurrent VUs

A vital technical distinction in load testing: **Virtual Users (VUs) represent concurrent network traffic threads, not distinct database records.**

- 1. Public / Anonymous Endpoints (0 database accounts needed):** Landing page (/), public course catalog, and certificate cryptographic verification (/verify/[code]) handle hundreds of concurrent requests without touching user sessions.
- 2. Round-Robin Session Distribution:** In k6, 500 virtual threads can simulate concurrent browsing by cycling across the 50 seeded database accounts (VU % 50), accurately stressing connection pools, auth token verification, and database query throughput.

6. Database Summary & Seeded User Directory

The ACCESS database is initialized with verified demo personas, structured curriculum, and complete learning histories. All demo accounts use the standard password demo1234.

6.1 Seeded User Directory & Scenario Roles

Full Name	Email Address	Role	Target Preset / Profile	Assigned Test Scenario
Aliff Affandi	aliff.admin@access.edu.my	admin	None (Standard)	Super Admin; full platform governance, analytics & user management
Nurul Izzah	nurul.admin@access.edu.my	admin	None (Standard)	Secondary Admin; course approvals & executive reporting
Dr. Sarah Chen	sarah.educator@access.edu.my	educator	None (Author)	Lead Educator; author of ADHD & Executive Function courses
Prof. James Wilson	james.educator@access.edu.my	educator	None (Author)	Special Education Lead; author of Dyslexia & Literacy courses
Amir Hakim	amir.learner@access.edu.my	learner	ADHD Preset	ADHD Learner; tests chunking, action-first banner, task checklist, video questions
Hafizuddin Danial	hafiz.learner@access.edu.my	learner	Dyslexia Preset	Dyslexic Learner; tests OpenDyslexic font, TTS Shift-reading, cream tint, reading ruler
Farah Nabilah	farah.learner@access.edu.my	learner	Autism Preset	Autistic Learner; tests sensory calm mode, muted pastels, visual schedule, summaries
David Lee	david.learner@access.edu.my	learner	Vision Preset	Low Vision; tests high-contrast, large text scaling, screen reader alt-text
Elena Rostova	elena.learner@access.edu.my	learner	General / Custom	Multi-device student; tests responsive layout, certificates & badges

6.2 Summary of Initialized Database Metrics

Database Entity	Metric Count	Database Entity	Metric Count
Public Database Tables	42 Relational Tables	Total Enrolled Tracks	21 Active Enrollments
System Admin Accounts	3 Verified Accounts	Lesson Progress Records	64 Progress Telemetry Rows
Educator Accounts	3 Verified Educators	Quiz Submissions & Attempts	19 Graded Attempts
Student Learner Personas	15 Seeded Learners	Interactive Video Questions	12 Checkpoint Questions
Published Courses	6 Multi-Chapter Courses	Earned Achievement Badges	38 Awarded Badges
Structured Lessons	24 Interactive Lessons	Issued Digital Certificates	14 Verified Credentials

REPORT APPROVAL & VERIFICATION STATUS:

This document confirms that the ACCESS platform architecture meets all specified functional requirements, WCAG 2.2 accessibility standards, and deterministic auditing benchmarks. All test procedures outlined in Section 5 can be reproduced against both local and production environments.

Evaluator / Developer Signature: _____ Date: _____